Article type: Review

Epidemiology of injuries in male and female youth football players: A systematic review and meta-analysis

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Supplementary materials

Supplementary File 1. PRISMA checklist.

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3–4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4 (Suppl 2)
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4–5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4–5 (Suppl 3)
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4–5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5 (Suppl 5)
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5 (Suppl 4)
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5–6 (Suppl 6 and 7)
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6–7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6–7

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6–7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7 (Fig 1)
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7 (Suppl 8)
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	7 (Suppl 9 and 10)
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8 (Fig 2–5)
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	8–11 (Fig 2–6)
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	7 (Suppl 11)
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	8–11
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12–16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	16–17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	17
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	17

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta–Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: **www.prisma–statement.org**.

Supplementary File 2. Inclusion/exclusion criteria for young football players' injuries literature search.

	Inclusion criteria	Exclusion criteria	Rationale for these criteria
Publication type	Peer-reviewed original research articles only.	Non-peer-reviewed articles, newspapers, opinion pieces, systematic reviews and meta-analysis, editorials, commentaries and letters to the editor. Conference proceedings/abstracts. Book chapters.	For reasons of practicality, it was deemed acceptable to include only studies published in peer-reviewed journals.
Language	English and Spanish	Non-English and Spanish.	For reasons of practicality, it was deemed acceptable to include only studies published in English or Spanish.
Publication date	Up to 31st December 2020.	–	All articles were included regardless of the time period.
Study design	Multi-centre studies, randomised control trials, and prospective cohort studies	Descriptive studies, anecdotal. Case–controlled studies and cross-sectional studies. Case studies or expert opinion.	Based on the evidence hierarchy as a guide, ONLY study designs ranked 'good' and 'excellent' were included in this systematic review and meta–analysis. This was to ensure high methodological rigour and offer reasonable empirical support for the incidence and aetiology of injuries among young footballers.
Sex and age	Male and female football players younger than 19 years old (U19).	Studies with no football players, where the football players are older than 19 years old or participants' age is not specified, and studies that do not report injury incidences separately by sex.	The primary outcome of interest was the incidence, nature and anatomical location of the injuries sustained in young male and female football players. Studies that compared injury rates between young and adult players and present injury incidences separately were included. Similarly, studies that analyse injury incidences in several sports including football but present injury incidence in young footballers separately were also included.
Playing level	Participating in elite and/or sub-elite level.	Non-competitive football activities.	The inclusion of injuries associated with such a range of different populations might result in a vague aetiology due to differences in

Supplementary File 2. Inclusion/exclusion criteria for young football players' injuries literature search.

	Inclusion criteria	Exclusion criteria	Rationale for these criteria
			playing/training conditions. Therefore, only studies with young football players participating in games/training in a competitive setting (elite and sub-elite) were included, to be consistent with the aims of this systematic review and meta-analysis. Elite involved professional youth academies, national teams and international tournaments. Players not described as belonging to a football academy, playing at a high level or classified as elite were considered as sub-elite (e.g., community, regional and inter-provincial playing levels).
Injury definition	Time-loss definition.	Non-Time-loss definition.	Time-loss injury, defined as any physical complaint sustained by a player that results from a football match or football training, irrespective of the need for medical attention, allows to compare statistically data between different studies reducing inconsistencies between data collection procedures. Studies with other definitions would not meet the methodological criteria of the meta-analysis.

Supplementary File 3. Search strategies.

Search strategy in MEDLINE/PubMed – 998 results

#1 (soccer[tiab] OR "soccer"[MeSH Terms] OR football[tiab] OR "football"[MeSH Terms]) AND (injury[tiab] OR "injury"[MeSH Terms] OR injuries[tiab] OR "injuries"[MeSH Terms] OR incidence[tiab] OR "incidence"[MeSH Terms] OR prevalence[tiab] OR "prevalence"[MeSH Terms] OR epidemiology[tiab] OR "epidemiology"[MeSH Terms]) AND (youth[tiab] OR "youth"[MeSH Terms] OR children[tiab] OR "children"[MeSH Terms] OR adolescents[tiab] OR "adolescents"[MeSH Terms] OR kids[tiab] OR "kids"[MeSH Terms])

#2 #1 Filters: Published up to 31st December 2020.

Search strategy in the Cochrane Central Register of Controlled Trials – 161 results

#1 soccer [Title/Abstract/Key Word] OR football [Title/Abstract/Key Word] AND injur [Title/Abstract/Key Word] OR incidence [Title/Abstract/Key Word] AND youth [Title/Abstract/Key Word]

#2 soccer [Title/Abstract/Key Word] OR football [Title/Abstract/Key Word] AND injur [Title/Abstract/Key Word] OR incidence [Title/Abstract/Key Word] AND children [Title/Abstract/Key Word]

#3 soccer [Title/Abstract/Key Word] OR football [Title/Abstract/Key Word] AND injur [Title/Abstract/Key Word] OR incidence [Title/Abstract/Key Word] AND adolescent [Title/Abstract/Key Word]

#4 #1 AND #2 AND #3

#5 #4 Filters: Published up to 31st December 2020.

Search strategy in Web of Science – 271 results

#1 TITLE: (football OR soccer) AND TITLE: (injur* OR incidence OR prevalence OR epidemiology) AND TITLE: (youth OR children OR adolescent OR kid)

Search strategy in Sportdiscus – 699 results

#1 AB (football OR soccer) AND AB (injury OR injuries OR incidence OR epidemiolog* OR prevalence) AND AB (youth OR children OR adolescent OR kid)

Supplementary File 4. Definitions used to include studies in the meta-analysis.

Term	Definition
Injury	Any physical complaint sustained by a player that results from a football match or football training, irrespective of the need for medical attention or time loss from football activities.
Time loss injury	Injury that results in a player being unable to take a full part in future football training or match play.
Recurrent injury	An injury of the same type and at the same site as an index injury and which occurs after a player's return to full participation from the index injury.
Injury severity	The number of days that have elapsed from the date of injury to the date of the player's return to full participation in team training and availability for match selection. Injuries are grouped as: *Minimal* Absence (1–3 days) *Minor / Mild* Absence (4–7 days) *Moderate* Absence (8–28 days) *Major / Severe* Absence (>28 days)
Match exposure	Play between teams from different clubs.
Training exposure	Team-based and individual physical activities under the control or guidance of the team's coaching or fitness staff that are aimed at maintaining or improving players' football skills or physical condition.
Overuse injury	An injury caused by repeated microtrauma without a single, identifiable event responsible for the injury.
Traumatic injury	Injury with sudden onset and known cause.
Contact injury	An injury caused by external influence (i.e., any contact with another player or other object).
Non–contact injury	An injury happened without external influence.
Injury location	▪ Head and neck (Head/face; Neck/cervical spine) ▪ Upper limbs (Shoulder/clavicula; Upper arm; Elbow; Forearm; Wrist; Hand/finger/thumb) ▪ Trunk (Sternum/ribs/upper back; Abdomen; Lower back/pelvis/sacrum) ▪ Lower limbs (Hip/groin; Thigh; Knee; Lower leg/Achilles tendon; Ankle; Foot/toe)
Type of injury grouping	• Fractures and bone stress • Joint (non-bone) and ligament (Dislocation/subluxation; Sprain/ligament injury; Lesion of meniscus or cartilage) • Muscle and tendon (Muscle rupture/tear/strain/cramps; Tendon injury/rupture/tendinosis/bursitis)

	• Contusions (Haematoma/contusion/bruise) • Laceration and skin lesion (Abrasion; Laceration) • Central/peripheral nervous system (Concussion [with or without loss of consciousness]; Nerve injury) • Other (Dental injuries; Other injuries)
Injury incidence	Number of injuries per 1000 player hours ((Σ injuries/Σ exposure hours) ×1000).

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Supplementary File 5. Moderator variables codded.

General study descriptors
▪ Authors
▪ Year of the study
▪ Continent / Tournament
▪ Sampling time (number of seasons)
Description of the study population
▪ Sample size
▪ Number of teams
▪ Age group
▪ Level of play (sub-elite or elite)
▪ Sex
Epidemiological descriptors
▪ Injury definition
▪ Number of injuries (total, match and training)
▪ Exposure time (total, match and training)
▪ Incidence (total, match and training)
▪ Injury burden or days lost per injury
▪ Injury location
▪ Type of injury
▪ Severity of injury
▪ Recurrence
▪ Injury mechanism (traumatic or overuse; contact or non–contact)
▪ Quality of the study (STROBE scale)
▪ Risk of bias (adapted NOS scale)

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76 **Supplementary File 6.** Description of the 22 STROBE criteria designed to assess quality of the studies included
77 in the meta–analysis.

	Item	Recommendation
Title and abstract	1	(*a*) Indicate the study's design with a commonly used term in the title or the abstract
		(*b*) Provide in the abstract an informative and balanced summary of what was done and what was found
Introduction		
Background/rationale	2	Explain the scientific background and rationale for the investigation being reported
Objectives	3	State specific objectives, including any prespecified hypotheses
Methods		
Study design	4	Present key elements of study design early in the paper
Setting	5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow–up, and data collection
Participants	6	(*a*) *Cohort study*—Give the eligibility criteria, and the sources and methods of selection of participants. Describe methods of follow-up
		(*b*) *Cohort study*—For matched studies, give matching criteria and number of exposed and unexposed
Variables	7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable
Data sources/ measurement	8*	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group
Bias	9	Describe any efforts to address potential sources of bias
Study size	10	Explain how the study size was arrived at
Quantitative variables	11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why
Statistical methods	12	(*a*) Describe all statistical methods, including those used to control for confounding
		(*b*) Describe any methods used to examine subgroups and interactions
		(*c*) Explain how missing data were addressed
		(*d*) *Cohort study*—If applicable, explain how loss to follow–up was addressed
		(*e*) Describe any sensitivity analyses
Results		
Participants	13*	(a) Report numbers of individuals at each stage of study-eg numbers potentially eligible, examined for eligibility, confirmed eligible, included

		in the study, completing follow-up, and analysed
		(b) Give reasons for non-participation at each stage
		(c) Consider use of a flow diagram
Descriptive data	14*	(a) Give characteristics of study participants (eg demographic, clinical, social) and information on exposures and potential confounders
		(b) Indicate number of participants with missing data for each variable of interest
		(c) *Cohort study*—Summarise follow-up time (eg, average and total amount)
Outcome data	15*	*Cohort study*—Report numbers of outcome events or summary measures over time
Main results	16	(*a*) Give unadjusted estimates and, if applicable, confounder–adjusted estimates and their precision (eg, 95% confidence interval). Make clear which confounders were adjusted for and why they were included
		(*b*) Report category boundaries when continuous variables were categorized
		(*c*) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period
Other analyses	17	Report other analyses done—eg analyses of subgroups and interactions, and sensitivity analyses
Discussion		
Key results	18	Summarise key results with reference to study objectives
Limitations	19	Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias
Interpretation	20	Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence
Generalisability	21	Discuss the generalisability (external validity) of the study results
Other information		
Funding	22	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based

Note: An Explanation and Elaboration article discusses each checklist item and gives methodological background and published examples of transparent reporting. The STROBE checklist is best used in conjunction with this article (freely available on the Web sites of PLoS Medicine at http://www.plosmedicine.org/, Annals of Internal Medicine at http://www.annals.org/, and Epidemiology at http://www.epidem.com/). Information on the STROBE Initiative is available at www.strobe–statement.org.

Supplementary File 7. Description of the 8 criteria designed to assess risk of bias of external validity quality in the studies[†]. This instrument is an adapted version of the Newcastle Ottawa Scale (NOS) for cohort studies.

Criterion	Description of criteria
1. Description or type of football players.	There are several types of football players (sub-elite *vs*. elite, males *vs*. females). Without the description regarding to the type of football players it is impossible to conclude which population the incidence rates refer to. Studies that reported a description of the football players or informed the type of football players receive a star for this criterion. Studies conducted in football tournaments (which may determine the type of football players; e.g., World Cup tournaments) and which describe the race characteristics receive a star for this criterion as well. Studies that did not describe the characteristics or the type of football players, and studies conducted in football tournaments that did not describe the characteristics of the tournament did not receive a star for this criterion.
2. Definition of football-related injury.	Studies that aimed to investigate football-related injuries should present a definition of an injury informing what was considered as an injury in the study. Studies that present a definition of time-loss injury received a star for this criterion.
3. Representativeness of the exposed cohort.	(a) Truly representative of the average football players in the community*; (b) somewhat representative of the average football players in the community*; (c) selected group of users; (d) no description of the derivation of the cohort.
4. Ascertainment of exposure.	(a) Secure record*; (b) structured interview*; (c) written self-report; (d) no description
5. Demonstration that outcome of interest was not present at start of study.	(a) Yes*; (b) no. Studies that described that all football players included were injury-free at baseline received a star for this criterion.
6. Assessment of outcome.	(a) Independent blind assessment*; (b) record linkage*; (c) self-report; (d) no description.
7. Was follow-up long enough for outcomes to occur risk factors.	(a) Yes*; (b) no. Studies that carried out a follow-up period of at least 12 weeks received a star for this criterion.
8. Adequacy of follow-up of cohorts	(a) Complete follow-up of all subjects accounted for*; (b) subjects lost to follow-up unlikely to introduce bias (up to 20 % loss) or description provided of those lost*; (c) follow-up rate <80% and no description of

	those lost; (d) no statement. A loss to follow–up greater than 20 % may increase the risk of bias in prospective studies (Fewtrell et al., 2008).

* Articles with this alternative received a star for this criterion.

†: The articles could be awarded a maximum of one star for each item. A total of 8 stars could be given for the articles.

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Supplementary File 8. Characteristics of the studies included in the meta-analysis.

Reference	Study	Participant			Exposure (hour)			Injury			Incidence			STROBE – /34	NOS – /8
Continent (or event) / Year / Level of play	duration (week)	Sex (cohort)	Age (year)	Teams (players)	Overall	Training	Match	Overall	Training	Match	Overall	Training	Match	(reporting quality)	(methodological quality)
Andreasen et al.[72]	1	M	U19 (10–19)	– (9586)	–	–	25527	–	–	92	–	–	3.6	16	5
IT / 1991 / EL		F	U19 (10–19)	– (3321)	–	–	8890	–	–	39	–	–	4.4		
Azuma et al.[47 CS] Control	40	M	U18 (15–18)	– (60)	25188	21408	3780	94	49	45	3.7	2.3	11.9	28	7
AS / 2018–19 / SEL															
Backous et al.[73]	3	M	U17 (6–17)	– (681)	14931.5	–	–	109	–	–	7.3	–	–	15	5
NA / – / MI		F	U17 (6–17)	– (458)	10094.3	–	–	107	–	–	10.6	–	–		
Bianco et al.[24 CS]	44	M (a)	U16 (13–16)	– (54)	59058	53616	5442	72	60	12	1.2	1.2	2.2	20	6
EU / 2012–13 / EL		M (b)	U19 (17–19)	– (23)	24302	21984	2318	35	25	10	1.4	1.1	4.3		
		M (T)	U19 (13–19)	– (80)	83360	75600	7760	107	85	22	1.3	1.2	2.8		
Brito et al.[34 CS]	6	M	U19 (12–19)	40 (741)	23364	20847	2517	53	37	16	2.5	1.8	6.8	21	5
EU / 2009 / SEL															
Brito et al.[33 CS]	43	M (a)	U12 (11–12)	– (179)	41666.7	–	–	25	–	–	0.6	–	–	22	7
EU / 2008–09 / SEL		M (b)	U14 (13–14)	– (169)	37272.7	–	–	41	–	–	1.1	–	–		
		M (c)	U16 (15–16)	– (165)	40714.3	–	–	57	–	–	1.4	–	–		
		M (d)	U18 (17–18)	– (161)	44705.9	–	–	76	–	–	1.7	–	–		
		M (T)	U18 (11–18)	28 (674)	161850	149803	12047	199	139	60	1.2	0.9	4.7		
Bult et al.[23 CS]	117	M (a)	U12 (U12)	– (17)	3583	–	–	21	–	–	5.9	–	–	26	7
EU / 2013–16 / EL		M (b)	U13 (U13)	– (50)	9965	–	–	51	–	–	5.1	–	–		
		M (c)	U14 (U14)	– (54)	11332	–	–	84	–	–	7.4	–	–		
		M (d)	U15 (U15)	– (54)	11175	–	–	139	–	–	12.4	–	–		
		M (e)	U16 (U16)	– (53)	13066	–	–	113	–	–	8.7	–	–		
		M (f)	U17 (U17)	– (38)	11761	–	–	119	–	–	10.1	–	–		
		M (g)	U19 (U19)	– (43)	13475	–	–	93	–	–	6.9	–	–		
		M (T)	U19 (U12–19)	– (170)	74358	–	–	620	–	–	8.3	–	–		

Cezarino et al.[22] CS SA / 2017 / EL	52	M (a)	U11 (10–11)	– (23)	4883.8	4516.7	367.1	2	1	1	0.4	0.2	2.7	29	8
		M (b)	U12 (11–12)	– (22)	4456.1	3908	548	8	8	0	1.8	2	0		
		M (c)	U13 (12–13)	– (25)	8120.4	7572.2	548.1	6	3	3	0.7	0.4	5.5		
		M (d)	U14 (13–14)	– (28)	12834.8	12394.8	440	21	17	4	1.6	1.4	9.1		
		M (e)	U15 (14–15)	– (28)	13176.4	12420.5	755.8	12	10	2	0.9	0.8	2.6		
		M (f)	U16 (15–16)	– (25)	12386.4	11731.2	655.2	27	24	3	2.2	2	4.6		
		M (g)	U17 (16–17)	– (28)	15084.9	14060.2	1024.7	46	32	14	3	2.3	13.7		
		M (h)	U18 (17–18)	– (16)	10359	9864	495	18	14	4	1.7	1.4	8.1		
		M (T)	U18 (10–18)	– (195)	81301.7	76467.8	4833.9	140	109	31	1.7	1.4	6.4		
Chena–Sinovas et al.[21] CS EU / – / SEL	40	M (a)	U9 (7–9)	– (68)	8337.4	7492.8	844.7	19	12	7	2.3	1.6	8.3	15	6
		M (b)	U11 (10–11)	– (80)	14830.3	13290.3	1540	23	17	6	1.6	1.3	3.9		
		M (c)	U13 (12–13)	– (114)	22518	19681.9	2836.2	55	52	3	2.4	2.6	1.1		
		M (d)	U15 (14–15)	– (71)	14973.8	12905.8	2068	69	41	28	4.6	3.2	13.5		
		M (e)	U18 (16–18)	– (69)	18121.4	16058.9	2062.5	102	63	39	5.6	3.9	18.9		
		M (T)	U18 (7–18)	– (402)	78781	69429.6	9351.3	268	185	83	3.4	2.7	8.9		
Clausen et al.[26] CS EU / 2012 / MI(T), EL(a), SEL(b)(c)	20	F (a)	U18 (15–18)	– (–)	6434	–	–	59	–	–	9.2	–	–	27	6
	20	F (b)	U18 (15–18)	– (–)	6811	–	–	63	–	–	9.2	–	–		
	20	F (c)	U18 (15–18)	– (–)	13761	–	–	140	–	–	10.2	–	–		
	20	F (T)	U18 (15–18)	32 (438)	27746	–	–	269	–	–	9.7	–	–		
Delecroix et al.[55] CS EU / 2013–17 / EL	156	M	U19 (16–19)	– (52)	23947.4	–	–	182	–	–	7.6	–	–	25	7
Ërgun et al.[31] CS EU / 2005–08 / EL	117	M	U19 (U17–19)	– (52)	2390.2	1897	493.2	29	14	15	12.1	7.4	30.4	21	6
Fouasson–Chailloux et al.[46] CS EU / 2011–16 / EL	195	M	U15 (13–15)	– (–)	44436	–	–	417	–	–	9.4	–	–	23	6
Frisch et al.[32] CS EU / 2007–08 / SEL	44	M	U19 (13–19)	– (67)	15673.1	12519.3	3153.7	163	89	74	10.4	7.1	23.5	27	7

Hägglund et al.[51] (a) [CS] EC / 2006 / EL	2	M	U19 (U19)	8 (144)	1253	762	490	8	0	8	6.4	0	16.3	25	6
Hägglund et al.[51] (b) [CS] EC / 2007 / EL	2	M	U19 (U19)	8 (147)	1158	654	504	15	1	14	13	1.5	27.8	25	6
Hägglund et al.[51] (c) [CS] EC / 2008 / EL	2	M	U19 (U19)	8 (145)	1461	957	504	13	2	11	8.9	2.1	21.8	25	6
Hägglund et al.[51] (d) [CS] EC / 2006 / EL	2	M	U17 (U17)	8 (144)	1316	834	482	11	1	10	8.4	1.2	20.7	25	6
Hägglund et al.[51] (e) [CS] EC / 2007 / EL	2	M	U17 (U17)	8 (145)	1161	685	477	7	1	6	6	1.5	12.6	25	6
Hägglund et al.[51] (f) [CS] EC / 2008 / EL	2	M	U17 (U17)	8 (144)	1354	899	455	18	5	13	13.3	5.6	28.6	25	6
Hägglund et al.[51] (g) [CS] EC / 2006 / EL	2	F	U19 (U19)	8 (144)	1707	1210	497	19	6	13	11.1	5	26.2	25	6
Hägglund et al.[51] (h) [CS] EC / 2007 / EL	2	F	U19 (U19)	8 (144)	1407	906	501	12	1	11	8.5	1.1	22	25	6
Hägglund et al.[51] (i) [CS] EC / 2008 / EL	2	F	U19 (U19)	8 (145)	1635	1121	514	8	2	6	4.9	1.8	11.7	25	6
Hawkins et al.[56] EU / 1994–97 / EL	117	M	U18 (16–18)	– (30)	16832.5	13902.4	2930.1	166	57	109	9.9	4.1	37.2	17	5
Imai et al.[54] [CS] Control AS / 2014–15 / SEL	32	M	U14 (12–14)	– (38)	7888	6126	1762	39	28	11	4.9	4.6	6.2	20	7
Junge et al.[74] (a) [CS] WC / 1999–2011 / EL	19	M	U17 (U17)	136 (2856)	–	–	9124.5	–	–	259	–	–	28.4	23	6
Junge et al.[74] (b) [CS] WC / 2008–12 / EL	9	F	U17 (U17)	48 (1008)	–	–	3168	–	–	68	–	–	21.5	23	6
Junge et al.[75] OC / 2001 / SEL	24	M	U18 (14–18)	12 (145)	9352.5	5727.5	3639.5	80	21	59	8.6	3.7	16.2	21	6

Junge et al.[76] Control EU / 1999–00 / SEL	52	M	U19 (14–19)	7 (93)	13094.4	–	–	100	–	–	7.6	–	–	22	7
Kakavelakis et al.[77] EU / 1999–00 / –	40	M	U15 (12–15)	24 (287)	52250	33333.3	17678.6	209	110	99	4	3.3	5.6	17	6
Kuzuhara et al.[43] AS / 2013–14 / SEL	52	M	U12 (≤12)	5 (86)	10838.4	8447.7	2390.8	25	12	13	2.3	1.4	5.4	22	7
		F	U12 (≤12)	– (3)	377.7	278.4	99.3	1	0	1	2.6	0	10.1		
Le Gall et al.[27 CS] EU / 1998–06 / EL	312	F	U19 (15–19)	– (119)	97325	87530	9795	619	400	219	6.4	4.6	22.4	24	7
Lislevand et al.[19 CS] AF / 2008 / SEL	0.3	F (a)	U13 (≤13)	37 (433)	–	–	431	–	–	5	–	–	11.6	27	5
		F (b)	U16 (13–16)	14 (213)	–	–	403	–	–	1	–	–	11.7		
		F (T)	U16 (≤16)	51 (646)	–	–	834	–	–	6	–	–	7.2		
Nielsen et al.[78] EU / 1986 / MI	44	M	U18 (16–18)	2 (30)	4554	3564	990	27	13	14	5.9	3.6	14.4	15	8
Nilsson et al.[20 CS] EU / 2013–14 / EL	88	M	U19 (15–19)	– (43)	10367	7678.6	1161.3	61	43	18	6.8	5.6	15.5	22	8
Nogueira et al.[25 CS] EU / 2015–16 / SEL	26	M (a)	U16 (15–16)	11 (290)	33673	28598.5	5074.5	138	73	65	3.7	2.1	12.6	29	7
		M (b)	U19 (17–19)	10 (239)	28389	24561	3828	110	46	64	4	2	16		
		M (T)	U19 (15–19)	21 (529)	62062	53159.5	8902.5	248	119	129	3.9	2.1	14.2		
Owoeye et al.[79 CS] Control AF / 2012–13 / EL	24	M	U19 (14–19)	10 (204)	61045	57448	3597	94	22	73	1.5	0.4	20.3	27	7
Raya–González et al.[18] EU / 2014–18/ EL	156	M (a)	U14 (13–14)	2 (–)	35064	31236	3828	84	61	23	2.4	2	6	23	7
		M (b)	U16 (15–16)	2 (–)	40300	35475	4825	111	67	44	2.8	1.9	9.1		
		M (c)	U19 (17–19)	2 (–)	49679	45318	4361	142	94	48	2.9	2.1	11		
		M (T)	U19 (13–19)	6 (257)	125043	112029	13014	337	222	115	2.7	2	8.8		
Renshaw et al.[9 CS] EU / 2012–13 / EL	39	M (a)	U11 (U9–11)	– (68)	11259.8	8695.7	2564.1	7	6	1	0.6	0.7	0.4	25	8
		M (b)	U15 (U15)	– (17)	97325	87530	150	–	–	12	–	–	80		
		M (c)	U16 (U16)	– (17)	–	–	343.8	–	–	11	–	–	32		
		M (d)	U18 (U18)	– (20)	–	2500	–	–	15	–	–	6	–		

		M (T)	U18 (9–18)	– (181)	29346	–	–	127	–	–	4.3	–	–		
Rommers et al.[48 CS] EU / 2017–18 / EL	39	M	U15 (U10–15)	– (734)	129206	112745	16464	389	229	160	3	2	9.7	20	7
Schmidt–Olsen et al.[80] IT / 1984 / EL	1	M (a)	U11 (9–11)	– (497)	–	–	1139.2	–	–	3	–	–	2.6	11	5
		M (b)	U13 (12–13)	– (1554)	–	–	3737.4	–	–	15	–	–	4		
		M (c)	U16 (14–16)	– (1932)	–	–	5729.2	–	–	45	–	–	7.8		
		M (d)	U19 (17–19)	– (1292)	–	–	3543.7	–	–	37	–	–	10.4		
		F (e)	U13 (9–13)	– (361)	–	–	13043.5	–	–	7	–	–	0.5		
		F (f)	U16 (14–16)	– (732)	–	–	1943	–	–	49	–	–	25.2		
		F (g)	U19 (17–19)	– (232)	–	–	635.6	–	–	13	–	–	20.9		
		M (T)	U19 (9–19)	– (5275)	–	–	14223.6	–	–	100	–	–	7.4		
		F (T)	U19 (9–19)	– (1325)	–	–	3913	–	–	69	–	–	17.6		
Sieland et al.[45] EU / 2015–17 / EL	78	M	U19 (U12–19)	– (205)	46296.3	40434.8	6400	125	93	32	2.7	2.3	5	24	7
Söderman et al.[35] EU / 1996 / SEL	28	F	U19 (14–19)	10 (153)	11689.2	–	–	79	–	–	6.8	–	–	20	7
Soligard et al.[53 CS] Control EU / 2007 / –	32	F	U17 (13–17)	– (837)	45428	31086	14342	215	74	138	4.7	2.4	9.6	32	8
Soligard et al.[44 CS] ET / 2005–08 / EL	4	M (a)	U13 (13)	– (–)	–	–	9095	–	–	20	–	–	2.2	24	5
		M (b)	U14 (14)	– (–)	–	–	12154	–	–	45	–	–	3.7		
		M (c)	U16 (15–16)	– (–)	–	–	16945	–	–	68	–	–	4		
		M (d)	U19 (17–19)	– (–)	–	–	6028	–	–	38	–	–	6.3		
		F (e)	U13 (13)	– (–)	–	–	2601	–	–	15	–	–	5.8		
		F (f)	U14 (14)	– (–)	–	–	4576	–	–	22	–	–	4.8		
		F (g)	U16 (15–16)	– (–)	–	–	8163	–	–	29	–	–	3.6		
		F (h)	U19 (17–19)	– (–)	–	–	3036	–	–	35	–	–	11.5		
		M (T)	U19 (13–19)	– (–)	–	–	44222	–	–	171	–	–	5.8		
		F (T)	U19 (13–19)	– (–)	–	–	18376	–	–	101	–	–	5.5		

Sprouse et al.[49 CS] EU / 2012–20 / EL	312	M (a)	U15 (U15)	– (–)	7958	7159	799	56	23	33	7	3.2	41.3	22	7
		M (b)	U16 (U16)	– (–)	9911	8435	1476	106	44	62	10.7	5.2	42		
		M (c)	U17 (U17)	– (–)	8702	6771	1931	65	20	45	7.5	2.9	23.3		
		M (d)	U18 (U18)	– (–)	6504	5332	1172	43	21	22	6.6	3.9	18.8		
		M (e)	U19 (U19)	– (–)	10689	9055	1634	49	25	24	4.6	2.8	14.7		
		F (f)	U15 (U15)	– (–)	7852	7531	321	61	52	9	7.8	6.9	28		
		F (g)	U16 (U16)	– (–)	7612	6633	979	48	28	20	6.3	4.2	20.4		
		F (h)	U17 (U17)	– (–)	15146	13651	1495	100	46	54	6.6	3.4	36.1		
		F (i)	U19 (U18–19)	– (–)	20541	18347	2194	117	67	50	5.7	3.6	22.8		
		M (T)	U19 (U15–19)	8 (–)	43764	36752	7012	319	133	186	7.3	3.6	26.5		
		F (T)	U19 (U15–19)	9 (–)	51151	46162	4989	326	193	133	6.4	4.2	26.7		
Steffen et al.[81 CS] Control EU / 2005 / –	32	F	U17 (13–17)	51 (947)	65725	–	–	241	–	–	3.7	–	–	29	7
Tears et al.[17 CS] EU / 2009–15 / EL	234	M	U18 (11–18)	– (–)	352800	–	–	778	–	–	2.2	–	–	25	6
van der Sluis et al.[10 CS] EU / 2002–07 / EL	117	M	U13 (–)	– (26)	33628.9	–	–	108	–	–	3.2	–	–	22	6
Wald én et al.[82 CS] EC / 2005 / EL	2	M	U19 (U19)	8 (144)	1394	899	495	17	2	15	13.4	2.9	30.4	25	6
Wik et al.[50 CS] AS / 2016–20 / EL	144	M (a)	U13 (U13)	– (–)	17072	15094	1978	133	91	42	7.8	6	21.2	28	7
		M (b)	U14 (U14)	– (–)	19245	16726	2519	164	105	59	8.5	6.3	23.4		
		M (c)	U15 (U15)	– (–)	17865	14803	3062	194	109	85	10.9	7.4	27.8		
		M (d)	U16 (U16)	– (–)	15719	12903	2816	215	114	101	13.7	8.8	35.9		
		M (e)	U17 (U17)	– (–)	13738	11203	2535	234	123	111	17	11	43.8		
		M (f)	U18 (U18)	– (–)	9188	7340	1848	171	97	74	18.6	13.2	40		
		M (T)	U18 (U12–18)	– (301)	92827	78069	14758	1111	639	472	12	8.2	32		
Zarei et al.[83 CS] Control AS / 2017 / EL	39	M	U14 (7–14)	17 (519)	32113	29716	2397	60	36	24	1.9	1.2	10	31	7

Notes: [CS] Study was implemented according to the 2006 consensus statement for epidemiological studies in football.

(a);(b);(c);(d);(e);(f);(g);(h);(i): indicate different cohorts in the same study; (T) indicates the total sample of the study.

Abbreviations: AF = Africa; AS =Asia; EC = European Championship; EL = Elite; ET = European Tournament; EU = Europe; F = Female; IT = International Tournament; M = Male; MI = Mixed (elite and sub–elite); NA = North America; NOS = Newcastle Ottawa Scale. OC = Oceania; SA = South America; SEL = Sub–elite; STROBE = Strengthening the Reporting of Observational Studies in Epidemiology statement; U = Under; WC = World Championship.

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Supplementary File 9. Analysis of the selected studies' methodological quality–STROBE (*n* = 43).

Study	1a	1b	2	3	4	5	6a	6b	7	8	9	10	11	12a	12b	12c	12d	12e	13a	13b	13c	14a	14b	14c	15	16a	16b	16c	17	18	19	20	21	22	Score
Andreasen et al.[72]	–	+	+	+	–	+	+	+	+	+	–	–	–	–	+	–	–	–	–	–	–	+	–	+	+	–	–	+	+	+	–	–	–	+	16
Azuma & Someya[47]	+	+	+	+	–	+	+	+	+	+	+	–	+	+	+	–	+	–	+	+	+	+	–	+	+	+	+	+	+	+	+	+	+	–	28
Backous et al.[73]	+	–	+	+	–	–	+	+	+	+	+	–	–	–	+	–	–	–	–	–	–	–	–	+	+	–	–	+	+	+	–	–	–	+	15
Bianco et al.[24]	+	+	+	+	–	+	+	+	+	+	+	–	–	–	+	–	–	–	–	–	–	+	–	+	+	–	–	+	+	+	+	+	–	+	20
Brito et al.[34]	+	+	–	+	+	+	–	+	+	+	–	–	+	–	+	–	–	–	+	–	–	–	–	+	+	+	+	+	+	+	+	+	–	+	21
Brito et al.[33]	+	+	+	+	–	+	–	+	+	+	–	–	+	–	+	–	–	–	+	+	–	+	–	+	+	+	+	+	+	+	–	+	–	+	22
Bult et al.[23]	+	+	+	+	–	+	+	+	+	+	–	–	+	+	+	–	–	–	+	–	+	+	–	+	+	+	+	+	+	+	+	+	+	+	26
Cezarino et al.[22]	+	+	+	+	+	+	+	+	+	+	+	–	+	+	+	+	+	–	+	–	–	+	+	+	+	+	+	+	+	+	+	+	–	+	29
Chena–Sinovas et al.[21]	–	+	+	+	–	–	–	+	+	+	+	–	+	–	+	–	–	–	–	–	–	+	–	+	+	–	–	+	+	–	–	+	–	–	15
Clausen et al.[26]	+	+	+	+	+	+	+	+	+	+	–	–	–	+	+	+	+	–	+	+	+	+	–	+	+	+	+	+	+	+	–	+	–	+	27
Delecroix et al.[55]	+	+	+	+	+	+	+	+	+	+	–	–	–	+	+	+	+	–	+	–	–	+	–	+	+	–	+	+	+	+	+	+	–	+	25
Ergün et al.[31]	+	+	+	+	–	+	+	+	+	+	–	–	+	–	+	–	+	–	–	–	–	+	–	+	+	–	–	+	+	+	+	+	–	+	21
Fouasson–Chailloux et al.[46]	+	+	+	+	–	+	+	+	+	+	+	–	+	–	+	–	–	–	–	–	–	+	–	+	+	+	+	+	+	+	+	+	–	+	23
Frisch et al.[32]	+	+	+	+	+	+	+	+	+	+	–	–	+	+	+	–	+	–	+	+	–	+	–	+	+	+	+	+	+	+	+	+	–	+	27
Hägglund et al.[51]	+	+	+	+	+	+	+	+	+	+	–	–	+	+	+	–	–	–	+	–	–	+	–	+	+	+	–	+	+	+	+	+	+	+	25
Hawkins & Fuller[56]	+	+	+	+	+	+	+	+	+	+	–	–	+	–	+	–	–	–	–	–	–	–	–	+	+	–	–	+	+	+	–	–	–	–	17
Imai et al.[54]	–	+	+	+	–	+	–	+	+	+	–	–	+	+	+	–	–	–	–	+	–	–	–	+	+	+	–	–	+	+	+	+	+	+	20
Junge & Dvorak[74]	+	+	+	+	–	+	+	+	+	+	–	–	+	+	–	–	–	–	+	–	–	+	+	+	+	+	+	–	+	+	–	+	+	+	23
Junge et al.[75]	+	+	+	–	+	+	+	+	+	+	–	–	–	–	+	+	+	–	+	+	–	+	–	+	+	–	–	+	+	–	+	+	–	–	21
Junge et al.[76]	+	+	+	+	+	+	+	+	+	+	–	–	–	–	+	+	+	–	+	+	–	+	–	+	+	–	–	+	+	–	–	+	–	+	22

Kakavelakis et al.[77]	+	+	+	+	–	+	+	–	+	+	–	–	–	–	–	–	+	–	+	+	–	+	–	+	+	–	–	+	–	+	–	+	–	–	17
Kuzuhara et al.[43]	+	+	+	+	–	+	+	+	+	+	–	–	–	+	+	–	–	–	–	–	–	+	–	+	+	+	+	+	+	+	+	+	–	+	22
Le Gall et al.[27]	+	+	+	+	+	+	+	+	+	+	+	–	–	–	+	–	+	–	–	–	–	+	–	+	+	+	+	+	+	+	+	+	–	+	24
Lislevand et al.[19]	+	+	+	+	+	+	+	+	+	+	+	–	+	+	+	–	–	–	–	–	–	+	+	+	+	+	+	+	+	+	+	+	+	+	27
Nielsen & Yde[78]	+	+	+	+	–	+	+	+	+	+	+	–	–	–	–	–	–	–	–	–	–	+	–	+	+	–	–	+	+	–	–	–	–	–	15
Nilsson et al.[20]	+	+	+	+	+	+	+	–	+	+	+	–	+	–	–	–	+	–	+	+	–	+	–	+	+	–	–	+	–	+	+	+	–	+	22
Nogueira et al.[25]	+	+	+	+	–	+	+	+	+	+	+	–	+	+	+	+	+	–	+	+	+	+	–	+	+	+	+	+	+	+	+	+	–	+	29
Owoeye et al.[79]	+	+	+	+	+	+	+	+	+	+	+	–	+	+	+	+	+	–	+	+	+	+	–	+	+	–	–	+	+	+	+	+	–	–	27
Raya–Gonz âlez et al.[18]	+	+	+	+	+	+	+	+	+	+	–	–	+	+	+	–	–	–	–	–	–	+	–	+	+	+	+	+	+	+	+	+	–	–	23
Renshaw & Goodwin[9]	+	+	+	+	+	+	+	+	+	+	+	–	+	+	+	–	+	–	–	–	–	–	–	+	+	–	+	+	+	+	+	+	+	+	25
Rommers et al.[48]	+	+	+	+	+	+	+	+	+	+	–	–	+	–	–	–	–	–	–	–	–	+	–	+	+	–	–	–	+	+	+	+	+	+	20
Schmidt–Olsen et al.[80]	–	–	+	+	–	+	–	+	+	+	–	–	–	–	–	–	–	–	–	–	–	+	–	+	+	–	–	+	+	–	–	–	–	–	11
Sieland et al.[45]	+	+	+	+	+	+	+	+	+	+	+	–	+	+	+	–	–	–	–	–	–	+	–	+	+	–	–	+	+	+	+	+	+	+	24
S öderman et al. [35]	+	+	+	+	–	+	+	+	+	+	+	–	–	–	–	–	+	–	+	+	–	+	–	+	+	–	–	+	+	–	–	+	–	+	20
Soligard et al.[53]	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	–	+	+	+	+	–	+	+	+	+	+	+	+	+	+	+	32
Soligard et al.[44]	+	+	+	+	+	+	+	+	+	+	+	–	+	+	+	–	–	–	–	–	–	–	+	–	+	+	–	+	+	+	+	+	+	–	+	24
Sprouse et al.[49]	–	+	+	+	+	+	+	+	+	+	–	+	+	+	+	–	–	–	–	–	–	–	–	–	+	+	–	–	+	+	+	+	+	+	+	22
Steffen et al.[81]	+	+	+	+	+	+	+	+	+	+	+	+	–	+	+	+	–	–	+	+	+	+	–	+	+	+	+	+	+	+	+	+	+	–	+	29
Tears et al.[17]	+	+	+	+	+	+	+	+	+	+	+	–	+	+	+	+	+	–	–	–	–	–	–	+	+	–	–	+	+	+	+	+	+	+	25
van der Sluis et al.[10]	–	+	+	+	–	+	+	+	+	+	–	–	+	–	+	+	+	–	+	+	–	–	+	+	+	–	–	+	+	+	+	+	–	–	22
Wald én et al.[82]	+	+	+	+	+	+	+	+	+	+	–	–	+	+	+	–	–	–	+	–	–	+	+	+	+	–	–	+	+	+	+	+	+	+	25

Wik et al.[50]	+	+	+	+	+	+	+	+	+	+	–	–	+	–	–	+	+	–	+	+	+	+	+	+	+	+	+	+	–	+	+	+	+	+	28
Zarei et al.[83]	+	+	+	+	+	+	+	+	+	+	–	+	+	+	+	+	+	+	+	+	+	+	+	–	+	+	+	+	–	+	+	+	+	+	31

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Supplementary File 10. Risk of bias assessment of the studies (Newcastle-Ottawa scale).

Study	Criteria for assessing risk of bias								Total
	1	2	3	4	5	6	7	8	
Andreasen et al.[72]	*	*	*			*		*	5
Azuma & Someya[47]	*	*	*		*	*	*	*	7
Backous et al.[73]	*	*	*			*		*	5
Bianco et al.[24]	*	*	*			*	*	*	6
Brito et al.[34]	*	*	*			*		*	5
Brito et al.[33]	*	*	*	*		*	*	*	7
Bult et al.[23]	*	*	*	*		*	*	*	7
Cezarino et al.[22]	*	*	*	*	*	*	*	*	8
Chena–Sinovas et al.[21]	*	*	*			*	*	*	6
Clausen et al.[26]	*	*	*			*	*	*	6
Delecroix et al.[55]	*	*	*		*	*	*	*	7
Ergün et al.[31]	*	*	*			*	*	*	6
Fouasson–Chailloux et al.[46]	*	*	*	*		*	*		6
Frisch et al.[32]	*	*	*	*		*	*	*	7
Hägglund et al.[51]	*	*	*	*		*		*	6
Hawkins & Fuller[56]		*	*			*	*	*	5
Imai et al.[54]	*	*	*	*		*	*	*	7
Junge & Dvorak[74]	*	*	*	*		*		*	6
Junge et al.[75]	*	*	*	*		*	*		6
Junge et al.[76]	*	*	*	*		*	*	*	7
Kakavelakis et al.[77]	*	*	*			*	*	*	6
Kuzuhara et al.[43]	*	*	*	*		*	*	*	7
Le Gall et al.[27]	*	*	*		*	*	*	*	7

Lislevand et al.[19]	*	*	*			*		*	5
Nielsen & Yde[78]	*	*	*	*	*	*	*	*	8
Nilsson et al.[20]	*	*	*	*	*	*	*	*	8
Nogueira et al.[25]	*	*	*	*		*	*	*	7
Owoeye et al.[79]	*	*	*	*		*	*	*	7
Raya–González et al.[18]	*	*	*	*		*	*	*	7
Renshaw & Goodwin[9]	*	*	*	*	*	*	*	*	8
Rommers et al.[48]	*	*	*	*	*	*	*		7
Schmidt–Olsen et al.[80]	*	*	*			*		*	5
Sieland et al.[45]	*	*	*	*	*	*	*		7
Söderman et al.[35]	*	*	*		*	*	*	*	7
Soligard et al.[53]	*	*	*	*	*	*	*	*	8
Soligard et al.[44]	*	*	*			*		*	5
Sprouse et al.[49]	*	*	*	*		*	*	*	7
Steffen et al.[81]	*	*	*		*	*	*	*	7
Tears et al.[17]	*	*	*			*	*	*	6
van der Sluis et al.[10]	*	*	*			*	*	*	6
Waldén et al.[82]	*	*	*	*		*		*	6
Wik et al.[50]	*	*	*	*		*	*	*	7
Zarei et al.[83]	*	*	*	*		*	*	*	7

Criteria for assessing risk of bias: (1) description or type of football players; (2) definition of injury; (3) representativeness of the exposed cohort; (4) ascertainment of exposure; (6) demonstration that outcome of interest was not present at start of study; (6) assessment of outcome; (7) was follow–up long enough for outcomes to occur; (8) adequacy of follow–up of cohorts.

* Star(s) awarded for each criterion.

Supplementary File 11. Summary of findings (GRADE)

N °of studies	Certainty assessment					Effect			Certainty
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	N °of events	N °of individuals	Incidence (95%CI)	
Overall injury incidence in male youth football players									
33 [9,10,17,18,20–25,31–34,43,45–51,54–56,73,75–79,82,83]	Observational studies	Serious [a]	Very serious [b]	Not serious [c]	Not serious	6873	7895	5.7 injuries per 1000h of exposure (4.5 to 6.9)	⊕⊕⊕◯ MODERATE
Training injury incidence in male youth football players									
25 [18,20–22,24,25,31–34,43,45,47–51,54,56,75,77–79,82,83]	Observational studies	Serious [a]	Very serious [b]	Not serious [c]	Not serious	2496	6692	2.8 injuries per 1000h of exposure (2.0 to 3.5)	⊕⊕⊕◯ MODERATE
Match injury incidence in male youth football players									
29 [18,20–22,24,25,31–34,43–45,47–51,54,56,72,74,75,77–80,82,83]	Observational studies	Serious [a]	Very serious [b]	Not serious [c]	Not serious	2559	24409	14.4 injuries per 1000h of exposure (11.0 to 17.8)	⊕⊕⊕◯ MODERATE
Overall injury incidence in female youth football players									
9 [26,27,35,43,49,51,53,73,81]	Observational studies	Serious [a]	Very serious [b]	Serious [d]	Not serious	1896	3388	6.8 injuries per 1000h of exposure (5.0 to 8.5)	⊕⊕◯◯ LOW

Training injury incidence in female youth football players									
5 [27,43,49,51,53]	Observational studies	Not serious	Very serious [b]	Serious [d]	Serious [e]	676	1392	2.6 injuries per 1000h of exposure (1.2 to 4.1)	⊕⊕◯◯ LOW
Match injury incidence in female youth football players									
10 [19,27,43,44,49,51,53,72,74,80]	Observational studies	Serious [a]	Very serious [b]	Serious [d]	Not serious	804	7692	15.0 injuries per 1000h of exposure (9.7 to 20.2)	⊕⊕◯◯ LOW

[a] Some studies[19,34,56,72,73,80] presented certain risk of bias (assessed with the Newcastle Ottawa Scale [NOS])

[b] High inconsistency ($I^2 > 90\%$)

[c] Sub–analyses by age–group and level of play are presented, so indirectness was not downgraded

[d] Not enough data to carry out sub–analyses by age groups (U12 and U16) and level of play

[e] Limited sample size

Supplementary File 12. Probabilities of injury for males and females over a typical season†

Reference	Age (range)	Match incidence	Match duration (min)	Average probability of injury to a player in a typical season (%)‡
	Males			
Azuma & Someya[47]	U18 (15–18)	11.9	90	41
Bianco et al.[24]	U16 (13–16)	2.2	80	8
	U19 (17–19)	4.3	90	18
	All groups (13–19)	2.8	80–90	11–12
Brito et al.[34]	U19 (12–19)	6.8	60–90	18–26
Brito et al.[33]	U18 (11–18)	4.7	60–90	13–19
Cezarino et al.[22]	U11 (10–11)	2.7	40	5
	U12 (11–12)	0.0	50	0
	U13 (12–13)	5.5	50	13
	U14 (13–14)	9.1	60	24
	U15 (14–15)	2.6	60	8
	U16 (15–16)	4.6	80	17
	U17 (16–17)	13.7	80	42
	U18 (17–18)	8.1	90	30
	All groups (10–18)	6.4	40–90	12–25
Chena–Sinovas et al.[21]	U9 (7–9)	8.3	40	15
	U11 (10–11)	3.9	60	11
	U13 (12–13)	1.1	70	4
	U15 (14–15)	13.5	80	42
	U18 (16–18)	18.9	90	57
	All groups (7–18)	8.9	40–90	16–33
Ërgun et al.[31]	U19 (U17–19)	30.4	80–90	70–75
H ägglund et al.[51] (a)	U19 (U19)	16.3	90	52
H ägglund et al.[51] (b)	U19 (U19)	27.8	90	71
H ägglund et al.[51] (c)	U19 (U19)	21.8	90	63
H ägglund et al.[51] (d)	U17 (U17)	20.7	80	56
H ägglund et al.[51] (e)	U17 (U17)	12.6	80	40
H ägglund et al.[51] (f)	U17 (U17)	28.6	80	68
Hawkins et al.[56]	U18 (16–18)	37.2	90	81
Junge & Dvorak[74] (a)	U17 (U17)	28.4	90	72
Junge et al.[75]	U18 (14–18)	16.2	90	52
Kuzuhara et al.[43]	U12 (<=12)	5.4	40	10
Nielsen et al.[78]	U18 (16–18)	14.4	90	48
Nilsson et al.[20]	U19 (15–19)	15.5	90	50
Nogueira et al.[25]	U16 (15–16)	12.6	80	40

	U19 (17–19)	16.0	90	51
	All groups (15–19)	14.2	80–90	43–47
Owoeye et al.[79]	U19 (14–19)	20.3	90	60
Raya–González et al.[18]	U14 (13–14)	6.0	80	21
	U16 (15–16)	9.1	90	34
	U19 (17–19)	11.0	90	39
	All groups (13–19)	8.8	80–90	30–33
Renshaw et al.[9]	U11 (U9–11)	0.4	40–60	1
	U15 (U15)	80.0	80	96
	U16 (U16)	32.0	80	72
Sieland et al.[45]	U19 (U12–19)	5.0	60–90	14–20
Soligard et al.[44]	U13 (13)	2.2	40	4
	U14 (14)	3.7	40	7
	U16 (15–16)	4.0	50	10
	U19 (17–19)	6.3	60	17
	All groups (13–19)	3.9	40–60	7–11
Sprouse et al.[49]	U15 (U15)	41.3	80	81
	U16 (U16)	42.0	80	81
	U17 (U17)	23.3	90	65
	U18 (U18)	18.8	90	57
	U19 (U19)	14.7	90	48
	All groups (U15–19)	26.5	80–90	65–70
Waldén et al.[82]	U19 (U19)	30.4	90	75
Wik et al.[50]	U13 (U13)	21.2	60	47
	U14 (U14)	23.4	70	56
	U15 (U15)	27.8	80	67
	U16 (U16)	35.9	90	80
	U17 (U17)	43.8	90	86
	U18 (U18)	40.0	90	83
	All groups (U13–18)	32.0	60–90	62–76
TOTAL	U12 and below (6 cohorts)		40–60	7 (0–15)[§]
	U13–U16 (21 cohorts)		40–90	39 (4–96)
	U17–U19 (23 cohorts)		60–90	56 (17–86)
	All age–groups (30 cohorts)		40–90	47 (1–96)
	Females			
Hägglund et al.[51] (g)	U19 (U19)	26.2	90	69
Hägglund et al.[51] (h)	U19 (U19)	22.0	90	63
Hägglund et al.[51] (i)	U19 (U19)	11.7	90	41
Junge & Dvorak[74] (b)	U17 (U17)	21.5	90	62
Kuzuhara et al.[43]	U12 (<=12)	10.1	40	18

Le Gall et al.[27]	U19 (15–19)	22.4	90	64
Lislevand et al.[19]	U16 (13–16)	11.7	50	25
	All groups (≤16)	7.2	30–50	10–16
Soligard et al.[53]	U17 (13–17)	9.6	40–50	18–21
Soligard et al.[44]	U13 (13)	5.8	40	11
	U14 (14)	4.8	40	9
	U16 (15–16)	3.6	50	8
	U19 (17–19)	11.5	60	29
	All groups (13–19)	5.5	40–60	10–15
Sprouse et al.[49]	U15 (U15)	28.0	80	67
	U16 (U16)	20.4	80	56
	U17 (U17)	36.1	90	80
	U19 (U18–19)	22.8	90	64
	All groups (U15–19)	26.7	80–90	66–70
TOTAL	U12 and below (1 cohort)		40	18
	U13–U16 (6 cohorts)		40–80	30 (8–67)
	U17–U19 (7 cohorts)		60–90	58 (29–80)
	All age–groups (10 cohorts)		30–90	43 (13–69)

† Only studies (and cohorts) with sufficient data to estimate the probability of injury are presented.

‡ Average probability of injury to a player using the Poisson distribution model of Parekh et al.[68] assuming a player plays a whole season of 30 matches with each match 40–90 min in duration (according to youth football regulations).

§ Total values are presented as means and ranges of probabilities.

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